

Table 2. Summary of Experimental Results of Series B

Run number	D_r (m)	H_0 (m)	L_0 (m)	T_a (s)	$\frac{U_f}{\sqrt{gD}}$	$\overline{v}_{fs}$ (m/s)	$\overline{v}_{int}$ (m/s)	$\overline{v}_{net}$ (m/s)	v_{Taylor} (m/s)	$\frac{D_r}{D}$	$\frac{V_{air}}{V_w}$	Geyser
Video camera												
B-1	0.016	0.66	0.61	8.48	0.423	0.975	1.450	0.475	0.131	0.32	9.03	Yes
B-10	0.016	0.77	0.61	8.68	0.413	0.796	1.189	0.393	0.131	0.32	7.74	Yes
B-11	0.016	0.88	0.61	8.4	0.437	0.567	1.012	0.445	0.131	0.32	6.77	Yes
B-12	0.016	0.66	1.2	6.68	0.410	1.152	1.455	0.304	0.131	0.32	17.76	Yes
B-13	0.016	0.66	1.8	4.68	0.403	1.583	2.018	0.435	0.131	0.32	26.63	Yes
B-14	0.031	0.66	0.61	8.44	0.425	0.306	0.522	0.215	0.190	0.62	2.40	No
B-23	0.031	0.77	0.61	8.36	0.429	0.116	0.323	0.207	0.190	0.62	2.06	No
B-24	0.031	0.88	0.61	8.32	0.431	0.073	0.283	0.209	0.190	0.62	1.80	No
B-25	0.031	0.66	1.2	6.52	0.421	0.579	0.810	0.231	0.190	0.62	4.73	Yes
B-26	0.031	0.66	1.8	4.72	0.399	0.515	0.750	0.235	0.190	0.62	7.09	Yes
B-32	0.041	0.66	0.61	8.32	0.431	0.290	0.529	0.239	0.216	0.82	1.37	No
B-41	0.041	0.77	0.61	8.32	0.431	0.219	0.462	0.242	0.216	0.82	1.18	No
B-42	0.041	0.88	0.61	8.28	0.433	0.120	0.354	0.234	0.216	0.82	1.03	No
B-43	0.041	0.66	1.2	6.6	0.415	0.368	0.614	0.246	0.216	0.82	2.70	No
B-44	0.041	0.66	1.8	4.68	0.403	0.323	0.561	0.239	0.216	0.82	4.06	No
High speed camera												
B-H1	0.016	0.66	0.61	8.07	0.444	0.924	1.231	0.301	0.131	0.32	9.03	Yes
B-H2	0.021	0.66	0.61	7.84	0.457	0.768	1.022	0.261	0.157	0.42	5.24	Yes
B-H3	0.026	0.66	0.61	8.18	0.438	0.657	0.916	0.267	0.174	0.52	3.42	Yes
B-H4	0.031	0.66	0.61	8.14	0.440	0.207	0.418	0.215	0.190	0.62	2.40	No
B-H5	0.036	0.66	0.61	8.1	0.443	0.261	0.481	0.228	0.205	0.72	1.78	No
B-H6	0.041	0.66	0.61	8.1	0.443	0.246	0.476	0.235	0.219	0.82	1.37	No
B-H7	0.046	0.66	0.61	8.22	0.436	0.203	0.441	0.244	0.232	0.92	1.09	No

Note: $\overline{v}_{fs}$ = average measured velocity of the free surface in vertical riser.

B-H6 model comparison added to the published Table 2 conditions

Source	Outcome	T_a (s)	$Y_{fs,max}$ (m)	$Y_{int,max}$ (m)	RMSE Y_{fs} (m)	RMSE Y_{int} (m)
B-H6 experiment	No	8.10	1.201	1.178	--	--
1D	No	8.66	0.786	0.700	0.310	0.317
OpenFOAM 2D	No	8.04	1.021	0.940	0.097	0.245